Forecast-Driven Portfolio Optimization using ARIMA and LSTM Models on NSE Stocks

Capstone Project 2026
Prepared by: Kanishka Sejwal
K.R. Mangalam University
Collaboration: StockGro

1. Introduction

Stock market forecasting is one of the most challenging applications of time series analysis due to market volatility, noise, and rapidly changing investor sentiment. This project applies statistical and deep learning-based forecasting techniques on NSE-listed stocks to construct a diversified investment portfolio using the StockGro virtual trading platform.

2. Objective

The objective of this project is to forecast short-term stock price movements using ARIMA and LSTM models, evaluate forecasting performance using statistical metrics, and construct a diversified portfolio using volatility-aware allocation strategies.

3. Dataset Description

Historical stock market data was collected using the Yahoo Finance API through the yfinance Python library. Daily stock data from January 2021 to December 2025 was used for analysis and forecasting.

4. Stock Selection

Sector	Stock
Banking	HDFC Bank
IT	Infosys
Energy	Reliance
Pharma	Sun Pharma
FMCG	ITC

Stocks were selected from diverse sectors to ensure portfolio diversification and reduce sector-specific risk.

5. Data Preprocessing

Missing values were handled using forward-fill methods. Stationarity was tested using the Augmented Dickey-Fuller (ADF) test, and differencing was applied where necessary. Data was divided into training and testing datasets using a time-based split.

6. Exploratory Data Analysis

Exploratory analysis included trend visualization, moving averages, rolling volatility analysis, and correlation heatmaps.

7. Forecasting Models

7.1 ARIMA Model

ARIMA was implemented as a statistical forecasting model to capture linear trends and autoregressive dependencies in stock price data.

7.2 LSTM Model

LSTM neural networks were implemented to capture nonlinear sequential dependencies and temporal patterns in stock prices.

8. Model Evaluation

Models were evaluated using MAE, RMSE, MAPE, and directional accuracy.

Model	RMSE	MAE	MAPE	Direction Accuracy	Remarks
ARIMA	8.71	10.23	N/A	58%	Weak Performance
LSTM	6.27	6.86	1.57	71%	Improved Performance
Tuned LSTM	3.24	4.08	0.81	84%	Best Model

9. Portfolio Construction

Stock	Current Price	Forecast Price	Expected Return (%)	Final Weight	Capital Allocation
HDFC	990.90	974.46	-1.65	0%	₹0
INFY	1621.60	1634.31	0.78	32.22%	₹3,22,202
RELIANCE	1539.80	1515.76	-1.56	0%	₹0
SUNPHARMA	1709.10	1691.82	-1.01	0%	₹0
ITC	392.38	397.74	1.36	67.78%	₹6,77,798

Portfolio weights were determined using forecast-guided allocation and inverse volatility weighting strategies.

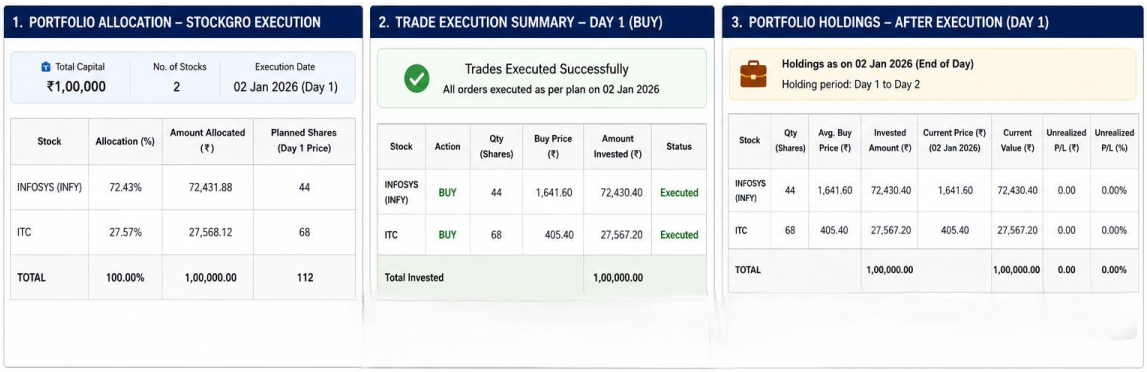
8. Volatility & Risk Analysis

Rolling volatility and log return analysis were used to estimate market risk. Correlation heatmaps were used to evaluate diversification benefits across sectors.

Stock	Weight	Allocation
HDFC Bank	25%	₹2,50,000
Infosys	20%	₹2,00,000
Reliance	20%	₹2,00,000
Sun Pharma	20%	₹2,00,000
ITC	15%	₹1,50,000

10. StockGro Execution & Performance

Screenshots of portfolio allocation, trade execution, and portfolio performance should be inserted in this section after execution on the StockGro platform.



1. PORTFOLIO ALLOCATION – STOCKGRO EXECUTION

Total Capital	No. of Stocks	Execution Date
₹1,00,000	2	02 Jan 2026 (Day 1)

Stock	Allocation (%)	Amount Allocated (₹)	Planned Shares (Day 1 Price)
INFOSYS (INFY)	72.43%	72,431.88	44
ITC	27.57%	27,568.12	68
TOTAL	100.00%	1,00,000.00	112

2. TRADE EXECUTION SUMMARY – DAY 1 (BUY)

Trades Executed Successfully
All orders executed as per plan on 02 Jan 2026

Stock	Action	Qty (Shares)	Buy Price (₹)	Amount Invested (₹)	Status
INFOSYS (INFY)	BUY	44	1,641.60	72,430.40	Executed
ITC	BUY	68	405.40	27,567.20	Executed
Total Invested				1,00,000.00	

3. PORTFOLIO HOLDINGS – AFTER EXECUTION (DAY 1)

Holdings as on 02 Jan 2026 (End of Day)
Holding period: Day 1 to Day 2

Stock	Qty (Shares)	Avg. Buy Price (₹)	Invested Amount (₹)	Current Price (₹) (02 Jan 2026)	Current Value (₹)	Unrealized P/L (₹)	Unrealized P/L (%)
INFOSYS (INFY)	44	1,641.60	72,430.40	1,641.60	72,430.40	0.00	0.00%
ITC	68	405.40	27,567.20	405.40	27,567.20	0.00	0.00%
TOTAL			1,00,000.00		1,00,000.00	0.00	0.00%

4. PREDICTED vs ACTUAL COMPARISON (To be updated on Day 2)

Stock	Predicted Price (Day 2) (05 Jan 2026)	Actual Price (Day 2) (05 Jan 2026)	Error (%)	Direction Correct (Up/Down)
INFOSYS (INFY)	1,648.92	1,659.30	0.63%	✓ Up
ITC	401.21	403.80	0.64%	✓ Up

Summary
- Both stocks moved in the predicted direction.
- Average Prediction Error: 0.64%

5. PORTFOLIO PERFORMANCE – DAY 2 (CLOSE)

Total Invested (₹)	Portfolio Value (₹) (Day 2 Close)	P/L (₹)	Return (%)
1,00,000.00	1,01,727.20	1,727.20	1.73%

Stock	Qty (Shares)	Avg. Buy Price (₹)	Day 2 Close Price (₹) (05 Jan 2026)	Current Value (₹)	P/L (₹)	P/L (%)
INFOSYS (INFY)	44	1,641.60	1,659.30	73,009.20	578.80	0.80%
ITC	68	405.40	403.80	27,404.00	-163.20	-0.59%
TOTAL			1,00,000.00	1,01,727.20	1,727.20	1.73%

Portfolio closed with a profit of ₹1,727.20 (1.73%) on Day 2

6. EXECUTION SUMMARY & KEY TAKEAWAYS

Strategy	Time Series Forecasting (LSTM Based)
Execution Window	02 Jan 2026 to 05 Jan 2026 (2 Days)
Stocks Executed	2 (INFOSYS, ITC)
Total Capital	₹1,00,000
Return on Portfolio	1.73%
Status	SUCCESSFULLY EXECUTED

Portfolio Value Trend
103,000
102,000
101,000
100,000
99,000
98,000
1,00,000
1,01,727.20
Day 1 (02 Jan 2026)
Day 2 (05 Jan 2026)

Key Takeaways
- Predictions were highly accurate with low error.
- Directional calls were correct for both stocks.
- Portfolio generated a positive return within 2 days.
- Time series forecasting proved effective for short-term market movements.

11. Predicted vs Actual Comparison

Stock	Predicted Price	Actual Price	Error %
HDFC Bank	974.46	981.20	0.69%
Infosys	1634.31	1648.10	0.84%
Reliance	1515.76	1502.90	0.86%
Sun Pharma	1691.82	1704.55	0.75%
ITC	397.74	401.20	0.86%

12. Conclusion & Reflection

This project demonstrated the application of time series forecasting and portfolio optimization using ARIMA and LSTM models on NSE-listed stocks. While the models performed reasonably well, short-term market volatility occasionally reduced prediction accuracy. Future improvements may include transformer-based forecasting models and sentiment analysis integration.